AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Major Project Synopsis

B. Tech (Artificial Intelligence)

Project Title: AI-Powered Chest X-ray Multi-Label Classification and Impression Generation Using Medical Vision and Language Models

Area: Artificial Intelligence / Multimodal Learning / Deep Learning

Academic Session: 2022-2026

Project Guide: Details of Project Team:

Programme: - Artificial Intelligence		Year/Semester: - 4th Year / 8th Semester	
S. No.	Enrollment No.	Name	Signature
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Abstract:

Chest X-rays (CXR) are one of the most commonly used diagnostic imaging tools, yet their interpretation requires specialized radiological expertise and is prone to delays, inconsistencies, and high workload. While deep learning models have achieved strong performance in multi-label disease classification, they often lack clinical explainability and fail to generate human-readable impressions similar to radiology reports.
This project proposes an AI-powered end-to-end system that integrates medical vision–language models, multimodal retrieval-augmented generation (RAG), and a healthy centroid–based contrastive guidance mechanism. The system performs multi-label pathology classification, retrieves clinically similar cases from large CXR archives, and generates accurate, evidence-grounded radiology impressions . Using BioMedCLIP for embedding generation and an LLM fine-tuned on medical report writing, the model reduces hallucination through contrastive retrieval and LLM-as-a-judge verification. This framework aims to support radiologists with faster, more consistent, and clinically meaningful diagnostic assistance.

Project Objectives: -

1. To perform multi-label classification of chest pathologies using medical vision–language embeddings (BioMedCLIP).
2. To develop a multimodal RAG pipeline that retrieves visually and clinically similar historical CXR cases.

3. To introduce a healthy baseline–guided contrastive retrieval mechanism that improves abnormality-focused case selection.
4. To generate clinically accurate radiology impressions using a fine-tuned medical LLM with minimal hallucination.
5. To build an integrated pipeline that provides classification, retrieval, and impression generation in a unified workflow.
6. To evaluate the system using standard medical datasets (MIMIC-CXR, IU-Xray) and ensure high factuality and clinical relevance.

Methodology to be adopted: -

1. Dataset Preparation
    1. Use MIMIC-CXR (377k images, 227k reports) and IU-Xray datasets.
    2. Clean text, extract findings and impression sections.
    3. Identify healthy cases using CheXpert labels to build a healthy baseline.
2. Multimodal Embedding Generation

    Use BioMedCLIP to produce:

    Image embeddings

    Report text embeddings

    Fuse into a single multimodal vector for each case.
3. Healthy Baseline Computation
    1. Compute a healthy centroid embedding from normal CXRs.
    2. Calculate abnormality scores as distance from this centroid.
4. RAG Store Design
    1. Store all multimodal embeddings + abnormality scores in FAISS (vector database).
    2. Enables fast similarity search.
5. Contrastive Retrieval

    Retrieve top similar images → re-rank using:

    $$\textbf{final_score} = \textbf{similarity} \times (1 + \lambda \times \textbf{case_abn}) \times (1 + \mu \times \textbf{query_abn})$$

    This prioritizes abnormal, clinically relevant matches.
6. Multi-Label Classification

    Classification head on BioMedCLIP embeddings predicts 14+ pathologies.
7. Impression Generation
    1. Fine-tune an LLM (Phi-3-Vision / LLaVA-Med) with LoRA
    2. Conditions:
        1. retrieved similar cases

2. classification predictions
3. abnormality deviation signals
3. Findings and Impression sections similar to real radiology reports.

8. Factuality Validation
    1. Secondary LLM acts as LLM-as-a-Judge.
    2. Checks hallucination and consistency with retrieved evidence.

Resource requirement (Hardware & software etc.): -

Hardware Requirements

1. GPU-enabled system (recommended: NVIDIA RTX 3090/4090 or A100)
2. Minimum:
    1. CPU: 8-core or above
    2. RAM: 16GB–32GB
    3. GPU VRAM: 12GB+
    4. Storage: 200GB (for datasets + embeddings)

Software Requirements

3. Programming: Python 3.9+
4. Libraries/Frameworks:
    1. PyTorch / TensorFlow
    2. Hugging Face Transformers
    3. FAISS (vector search)
    4. BioMedCLIP pretrained weights
    5. LoRA / PEFT for LLM fine-tuning

Justification of the project: -

Radiologist shortage and high workload lead to reporting delays.
Current AI systems provide labels, not clinically coherent impressions.
Lack of explainability makes current models hard to trust in hospitals.
Retrieval of similar cases is central to radiologist workflow but missing in most AI models.
Existing systems often hallucinate when generating reports.
Our system uniquely integrates:

1. Vision-language embeddings
2. Contrastive abnormality guidance
3. Evidence-grounded RAG
4. LLM-based impression generation with hallucination checks

Research Gap of work in minor project: -

- Pipeline was too heavy and multi-stage, making end-to-end tuning and optimization infeasible.
- Vision–language fusion lacked a validated mechanism for aligning global and local features.
- RAG retrieval was basic, lacking domain-optimized retrievers or evidence filtering for radiology.
- Generated impressions had limited grounding, as BioGPT was not fine-tuned on the target dataset.
- No healthy baseline or abnormality-aware retrieval, reducing the clinical relevance of retrieved cases.
- Retrieval ranking depended only on similarity, not on disease severity or deviation from normal.
- LLM lacked a justification module, making output correctness hard to verify.

Solutions/Mitigations Proposed in major project: -

1. Use BioMedCLIP multimodal embeddings to understand both CXR images and reports.
2. Build a healthy baseline (centroid) to detect and emphasize abnormalities.
3. Implement contrastive retrieval that re-ranks similar cases based on abnormality deviation.
4. Create a FAISS vector database for fast, accurate case retrieval.
5. Use a multi-label classification head to predict multiple thoracic diseases simultaneously.
6. Fine-tune a medical LLM (Phi-3 / LLaVA-Med) to generate radiology-style findings and impressions.
7. Add an LLM-as-a-judge module to check for hallucinations and ensure factual accuracy.
8. Integrate all components into one end-to-end diagnostic support system for radiologists.

This ensures higher task accuracy, generalization, and resilience against structural variations.

PERT Chart/Schedule of project completion: -

S. No.	Milestones	Number of weeks														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Project Initiation and Planning															
2	Literature Review + Gap Analysis															
3	Dataset access and Setup															
4	Data Cleaning and Preprocessing															
5	Multimodal Embedding Pipeline Development															
6	Contrastive Fine-Tuning / Metric Learning + Centroid & Abnormality Scoring Framework															
7	Vector Indexing, FAISS Optimization, Retrieval Pipeline & Classifier Integration															
8	LLM Fine-Tuning + RAG Integration + Explanation Generation Pipeline															
9	Evaluation and Optimization															
10	Research paper, documentation, Plagiarism Check and Editing															

References: Research papers : -

[1] CXR-RePaiR (2021)
[2] X-TRA: Cross-Modal Retrieval Augmentation for CXR (2023)
[3] CXR-CLIP / BioMedCLIP (2023–2024)
[4] LaB-RAG (2024)
[5] CXR-LLAVA (2023)
[6] MedCLIP (2022–2023)

Signature(s) of project Date:	Name and Signature of project guides